

Unit – 4 : Session – 11 : SLO – 2 | Converting a TCP Client to UDP

I. Key Changes (TCP client → UDP client)

1. Socket type: `socket.SOCK_STREAM` → `socket.SOCK_DGRAM`.
2. No connection setup: remove `connect()` — UDP is connectionless.
3. Sending: `send()/sendall()` → `sendto(data, (host, port))` — every datagram carries its destination.
4. Receiving: `recv()` → `recvfrom(bufsize)` — returns (data, address) because the sender must be identified.
5. No `listen()/accept()` on the peer; a UDP "server" just `bind()`s and calls `recvfrom()`.
6. Error handling: no automatic retransmission — timeouts and retries must be implemented by the application.

II. Example — UDP Client

1. `import socket`
2. `s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)`
3. `s.sendto(b"Hello", ("localhost", 5000))`
4. `data, addr = s.recvfrom(1024)`
5. `print(data); s.close()`